

Maths Final Exam Questions

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1 MCQ Part

1.1 Linear Algebra

Rules:

[Linear Algebra Cheat Sheet \(LinkedIn\)](#)

Introduction to matrices

[Google Drive](#)

Algebraic operations on matrices

[Google Drive](#)

Application on Inverse of Matrix

[Google Drive](#)

Gauss Elimination Method

[Google Drive](#)

1.2 SVD (Singular Value Decomposition)

1.2.1 Rules / Summary

- Any matrix A can be decomposed as:

$$A = U\Sigma V^T$$

- Where:

- U = orthogonal matrix (left singular vectors)
- Σ = diagonal matrix (singular values)
- V = orthogonal matrix (right singular vectors)

- Steps to compute SVD:

1. Compute:

$$A^T A$$

2. Find eigenvalues and eigenvectors of $A^T A$

- Eigenvalues $\rightarrow \lambda$
- Singular values:

$$\sigma_i = \sqrt{\lambda_i}$$

3. Construct V from eigenvectors of $A^T A$

4. Compute U using:

$$u_i = \frac{1}{\sigma_i} A v_i$$

5. Form Σ (diagonal matrix of σ_i)

- Rank of matrix = number of non-zero singular values

1.2.2 Hints

- Always start with $A^T A$ (easier for eigenvalues).
- Singular values are always **non-negative**.
- Sort singular values descending.
- Columns of V are eigenvectors of $A^T A$.
- Columns of U come from Av_i .
- If $\sigma = 0$, skip that vector when forming U .

1.2.3 Common Mistakes

- Forgetting square root: $\sigma \neq \lambda$
- Not normalizing eigenvectors
- Mixing U and V
- Wrong order of singular values
- Forgetting transpose in V^T
- Dividing by zero when $\sigma = 0$

1.2.4 Matrix A

Given:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Step 1: Compute $A^T A$

Since A is symmetric:

$$A^T A = A^2$$

$$A^T A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$$

Step 2: Compute Eigenvalues

$$\det(A^T A - \lambda I) = \begin{vmatrix} 10 - \lambda & 6 \\ 6 & 10 - \lambda \end{vmatrix} = 0$$

$$(10 - \lambda)^2 - 36 = 0$$

$$\lambda^2 - 20\lambda + 64 = 0$$

$$\lambda_1 = 16, \quad \lambda_2 = 4$$

Step 3: Singular Values

$$\sigma_1 = \sqrt{16} = 4, \quad \sigma_2 = \sqrt{4} = 2$$

Step 4: Eigenvectors (V)

For $\lambda_1 = 16$:

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 4$:

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 5: Compute U

$$u_i = \frac{1}{\sigma_i} A v_i$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 6: Final SVD

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A = U \Sigma V^T$$

1.2.5 Matrix B

Given:

$$B = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

Step 1: Compute $B^T B$

$$B^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$B^T B = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}$$

Step 2: Eigenvalues

$$\det(B^T B - \lambda I) = (11 - \lambda)^2 - 1 = 0$$

$$\lambda_1 = 12, \quad \lambda_2 = 10$$

Step 3: Singular Values

$$\sigma_1 = \sqrt{12}, \quad \sigma_2 = \sqrt{10}$$

Step 4: Eigenvectors (V)

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 5: Compute U

$$u_i = \frac{1}{\sigma_i} B v_i$$

Step 6: Final SVD

$$\Sigma = \begin{bmatrix} \sqrt{12} & 0 \\ 0 & \sqrt{10} \\ 0 & 0 \end{bmatrix}$$

$$B = U \Sigma V^T$$

1.2.6 Matrix C

Given:

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 1: Observe structure

Matrix is already diagonal (rectangular).

Step 2: Singular Values

They are absolute diagonal values:

$$\sigma_1 = 2, \quad \sigma_2 = 2, \quad \sigma_3 = 1$$

Step 3: Matrices

$$U = I_4, \quad V = I_3$$

Step 4: Final SVD

$$\Sigma = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$C = U \Sigma V^T$$

1.2.7 Matrix D

Given:

$$D = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$

Step 1: Observe structure

Columns are linearly dependent:

$$\text{col}_2 = 2 \cdot \text{col}_1$$

So rank = 1

Step 2: Compute $D^T D$

$$D^T D = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}$$

Step 3: Eigenvalues

$$\lambda_1 = 70, \quad \lambda_2 = 0$$

Step 4: Singular Values

$$\sigma_1 = \sqrt{70}, \quad \sigma_2 = 0$$

Step 5: Eigenvectors

$$v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Step 6: Compute U

$$u_1 = \frac{1}{\sqrt{70}} D v_1$$

Step 7: Final SVD

$$\Sigma = \begin{bmatrix} \sqrt{70} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

1.2.8 Question 2

Given:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Step 1: Compute $A^T A$

$$A^T A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Step 2: Compute AA^T

$$AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Step 3: Eigenvalues of AA^T

$$\lambda_1 = 3, \quad \lambda_2 = 1$$

Step 4: Singular Values

$$\sigma_1 = \sqrt{3}, \quad \sigma_2 = 1$$

Step 5: Compute U

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 6: Compute V

Solve eigenvectors of $A^T A$

Step 7: Final SVD

$$A = U\Sigma V^T$$

Conclusion:

- Singular values show scaling effect
- Larger $\sigma \rightarrow$ more important direction

1.3 PCA (Principal Component Analysis)

1.3.1 Rules / Summary

- PCA is used for **dimensionality reduction** while preserving maximum variance.
- Steps:

1. Compute mean: μ
2. Center data: $X_c = X - \mu$
3. Compute covariance matrix:

$$C = \frac{1}{n-1} X_c^T X_c$$

4. Compute eigenvalues λ and eigenvectors v
5. Sort eigenvalues descending:
 - Largest $\lambda \rightarrow$ PC1
 - Second $\lambda \rightarrow$ PC2

6. Project data:

$$z = X_c \cdot v$$

- Explained Variance Ratio:

$$EVR = \frac{\lambda_i}{\sum \lambda}$$

- PC1 captures the **maximum variance**.

1.3.2 Hints

- Always **center the data first**.
- If features have different scales $\rightarrow$ standardize.
- If matrix is:

$$\begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

then:

- Eigenvalues: $a + b, a - b$
- Eigenvectors: $[1, 1], [1, -1]$

- Eigenvectors must be **normalized**.
- Sign of eigenvector does not matter.

1.3.3 Common Mistakes

- Forgetting to subtract the mean.
- Using n instead of $(n - 1)$ in covariance.
- Not sorting eigenvalues.
- Forgetting normalization of eigenvectors.
- Confusing projection formula.
- Ignoring explained variance when choosing components.

1.3.4 Example 1 from sheet

Given:

$$X = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 4 \\ 5 & 5 \end{bmatrix}$$

Step 1: Compute the Mean Vector

$$\mu_1 = \frac{2+3+4+5}{4} = 3.5, \quad \mu_2 = \frac{1+2+4+5}{4} = 3$$

$$\mu = [3.5, 3]$$

Step 2: Center the Data

$$X_c = X - \mu = \begin{bmatrix} 2-3.5 & 1-3 \\ 3-3.5 & 2-3 \\ 4-3.5 & 4-3 \\ 5-3.5 & 5-3 \end{bmatrix} = \begin{bmatrix} -1.5 & -2 \\ -0.5 & -1 \\ 0.5 & 1 \\ 1.5 & 2 \end{bmatrix}$$

Step 3: Compute the Covariance Matrix

$$C = \frac{1}{n-1} X_c^T X_c$$

$$= \frac{1}{3} \begin{bmatrix} (-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2 & (-1.5)(-2) + (-0.5)(-1) + (0.5)(1) + (1.5)(2) \\ (-1.5)(-2) + (-0.5)(-1) + (0.5)(1) + (1.5)(2) & (-2)^2 + (-1)^2 + 1^2 + 2^2 \end{bmatrix}$$

$$C = \frac{1}{3} \begin{bmatrix} 5 & 7 \\ 7 & 10 \end{bmatrix} = \begin{bmatrix} 1.6667 & 2.3333 \\ 2.3333 & 3.3333 \end{bmatrix}$$

Step 4: Compute Eigenvalues

Solve:

$$\det(C - \lambda I) = 0$$

$$\begin{vmatrix} 1.6667 - \lambda & 2.3333 \\ 2.3333 & 3.3333 - \lambda \end{vmatrix} = 0$$

$$\lambda_1 \approx 4.98, \quad \lambda_2 \approx 0.02$$

Step 5: Compute Eigenvectors

For λ_1 :

$$v_1 = \begin{bmatrix} 0.5760 \\ 0.8174 \end{bmatrix}$$

For λ_2 :

$$v_2 = \begin{bmatrix} -0.8174 \\ 0.5760 \end{bmatrix}$$

Step 6: Project Data onto PC1

$$z_1 = X_c v_1$$

$$z_1 = \begin{bmatrix} -2.4989 \\ -1.1054 \\ 1.1054 \\ 2.4989 \end{bmatrix}$$

Conclusion:

- Variance explained by PC1:

$$\frac{4.98}{4.98 + 0.02} \approx 99.5\%$$

- Data can be reduced from 2D $\rightarrow$ 1D

1.3.5 Example 2 from sheet

Given:

$$X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{bmatrix}$$

Step 1: Compute the Mean Vector

$$\mu_1 = \frac{2 + 1 - 1 - 2}{4} = 0, \quad \mu_2 = \frac{1 + 2 - 2 - 1}{4} = 0$$

$$\mu = [0, 0]$$

Step 2: Center the Data

Since the mean is zero:

$$X_c = X - \mu = X$$

Step 3: Compute $X^T X$

$$X^T = \begin{bmatrix} 2 & 1 & -1 & -2 \\ 1 & 2 & -2 & -1 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 2 & 1 & -1 & -2 \\ 1 & 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{bmatrix}$$

Thus:

$$X^T X = \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix}$$

Step 4: Compute Covariance Matrix

$$C = \frac{1}{n-1} X^T X = \frac{1}{3} \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} & \frac{8}{3} \\ \frac{8}{3} & \frac{10}{3} \end{bmatrix}$$

Step 5: Compute Eigenvalues

Solve:

$$\det(C - \lambda I) = 0$$

$$\begin{vmatrix} \frac{10}{3} - \lambda & \frac{8}{3} \\ \frac{8}{3} & \frac{10}{3} - \lambda \end{vmatrix} = 0$$

$$\left(\frac{10}{3} - \lambda\right)^2 - \left(\frac{8}{3}\right)^2 = 0$$

$$\frac{100}{9} - \frac{20}{3}\lambda + \lambda^2 - \frac{64}{9} = 0$$

$$\lambda^2 - \frac{20}{3}\lambda + \frac{36}{9} = 0$$

$$\lambda^2 - \frac{20}{3}\lambda + 4 = 0$$

Solving:

$$\lambda_1 = 6, \quad \lambda_2 = \frac{2}{3}$$

Step 6: Compute Eigenvectors

For $\lambda_1 = 6$:

Solve $(C - 6I)v = 0$:

$$\begin{bmatrix} -\frac{8}{3} & \frac{8}{3} \\ \frac{8}{3} & -\frac{8}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$v_1 = v_2$$

Normalized:

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = \frac{2}{3}$:

$$v_1 = -v_2$$

Normalized:

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 7: Projection Formulas

$$z_1 = v_1^T x = \frac{x_1 + x_2}{\sqrt{2}}, \quad z_2 = v_2^T x = \frac{x_1 - x_2}{\sqrt{2}}$$

Step 8: Compute Scores

$$Z = XV$$

For first row $(2, 1)$:

$$z_1 = \frac{2 + 1}{\sqrt{2}} = \frac{3}{\sqrt{2}} \approx 2.1213$$

$$z_2 = \frac{2 - 1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.7071$$

Similarly computing all rows:

$$Z = \begin{bmatrix} 2.1213 & 0.7071 \\ 2.1213 & -0.7071 \\ -2.1213 & 0.7071 \\ -2.1213 & -0.7071 \end{bmatrix}$$

Conclusion:

- Variance explained by PC1:

$$\frac{6}{6 + \frac{2}{3}} = 0.9 = 90\%$$

- PC1 captures the direction $(x_1 + x_2)$
- PC2 captures the contrast $(x_1 - x_2)$

1.4 probability and z-distribution

1.4.1 Key Rules (Easy to Memorize)

1) Total Probability Rule

Discrete:

$$\sum p(x) = 1$$

Continuous:

$$\int f(x) dx = 1$$

—

2) Expected Value

Discrete:

$$\mathbb{E}[X] = \sum x p(x)$$

Continuous:

$$\mathbb{E}[X] = \int x f(x) dx$$

—

3) Variance

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

—

4) Expectation Tricks

$$\mathbb{E}[X^2] = \sum x^2 p(x)$$

$$\mathbb{E}[(X - a)^2] = \mathbb{E}[X^2] - 2a\mathbb{E}[X] + a^2$$

5) Linear Transformations

For constants a, b :

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Std}(aX + b) = |a| \text{Std}(X)$$

$$\text{Var}(X + b) = \text{Var}(X)$$

1.4.2 Binomial Distribution

Definition:

$$X \sim \text{Binomial}(n, p)$$

PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Mean and Variance:

$$\mu = np$$

$$\text{Var}(X) = np(1 - p)$$

Important Trick:

If asked about **non-defective** instead of defective:

$$p_{\text{new}} = 1 - p$$

1.4.3 Continuous Distributions

PDF Conditions:

- $f(x) \geq 0$
 - Total area = 1
-

Probability from PDF:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

1.4.4 Normal (Z) Distribution

Standardization Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Key Idea:

Convert ANY normal problem $\rightarrow$ Z-table

Important Values:

- $P(Z < 0) = 0.5$
- Symmetry: $P(Z < -a) = 1 - P(Z < a)$

Probability Tricks:

$$P(a < X < b) = P(X < b) - P(X < a)$$

$$P(X > a) = 1 - P(X < a)$$

1.4.5 VERY IMPORTANT HINTS

- Always check: **sum or integral = 1**
- In discrete $\rightarrow$ use summation, in continuous $\rightarrow$ use integration
- If you see $x^2 p(x) \rightarrow$ they want $\mathbb{E}[X^2]$
- Variance NEVER negative
- Binomial $\rightarrow$ memorize np and $np(1 - p)$ (comes directly)
- In normal:
 - Convert to Z first
 - Use table carefully (left vs right)
- If probabilities are given $\rightarrow$ think **reverse Z (solve for μ, σ)**
- Complement rule is VERY useful:

$$P(X \leq 1) = P(0) + P(1)$$

- In continuous problems:
 - First find constant k
 - THEN compute probability

1.4.6 First Quiz questions

In a machine learning monitoring system, let X denote the number of high-priority anomalies detected in a batch. Suppose X follows the probability mass function:

$$p(x) = cx^2, \quad x = 1, 2, 3$$

A. $c = \frac{1}{14}$, $\mathbb{E}[X] = \frac{18}{7}$, $\text{Var}(X) = \frac{19}{49}$, $\mathbb{E}[(X - 1)^2] = \frac{58}{7}$

B. $c = \frac{1}{2}$, $\mathbb{E}[X] = \frac{9}{4}$, $\text{Var}(X) = \frac{15}{16}$, $\mathbb{E}[(X - 1)^2] = \frac{29}{4}$

C. $c = \frac{1}{7}$, $\mathbb{E}[X] = \frac{16}{7}$, $\text{Var}(X) = \frac{18}{49}$, $\mathbb{E}[(X - 1)^2] = \frac{48}{7}$

D. $c = \frac{1}{10}$, $\mathbb{E}[X] = \frac{11}{5}$, $\text{Var}(X) = \frac{14}{25}$, $\mathbb{E}[(X - 1)^2] = \frac{24}{5}$

Step 1: Find c

We know that the total probability must equal 1:

$$\sum p(x) = 1$$

$$c(1^2 + 2^2 + 3^2) = 1$$

$$c(1 + 4 + 9) = 1$$

$$14c = 1 \Rightarrow c = \frac{1}{14}$$

Step 2: Find $\mathbb{E}[X]$ (Expected Value)

Formula:

$$\begin{aligned}\mathbb{E}[X] &= \sum x \cdot p(x) \\ &= \sum x \cdot (cx^2) = c \sum x^3 \\ &= \frac{1}{14}(1^3 + 2^3 + 3^3) \\ &= \frac{1}{14}(1 + 8 + 27) \\ &= \frac{36}{14} = \frac{18}{7}\end{aligned}$$

Step 3: Find $\mathbb{E}[X^2]$

$$\begin{aligned}\mathbb{E}[X^2] &= \sum x^2 \cdot p(x) \\ &= \sum x^2 \cdot (cx^2) = c \sum x^4\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{14}(1^4 + 2^4 + 3^4) \\
&= \frac{1}{14}(1 + 16 + 81) \\
&= \frac{98}{14} = 7
\end{aligned}$$

Step 4: Find Variance

Formula:

$$\begin{aligned}
\text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\
&= 7 - \left(\frac{18}{7}\right)^2 \\
&= 7 - \frac{324}{49} \\
&= \frac{343}{49} - \frac{324}{49} \\
&= \frac{19}{49}
\end{aligned}$$

Step 5: Find $\mathbb{E}[(X - 1)^2]$

Expand:

$$(X - 1)^2 = X^2 - 2X + 1$$

Take expectation:

$$\begin{aligned}
\mathbb{E}[(X - 1)^2] &= \mathbb{E}[X^2] - 2\mathbb{E}[X] + 1 \\
&= 7 - 2\left(\frac{18}{7}\right) + 1 \\
&= 7 - \frac{36}{7} + 1 \\
&= 8 - \frac{36}{7} \\
&= \frac{56}{7} - \frac{36}{7} \\
&= \frac{20}{7}
\end{aligned}$$

Final Answers:

$$c = \frac{1}{14}, \quad \mathbb{E}[X] = \frac{18}{7}, \quad \text{Var}(X) = \frac{19}{49}, \quad \mathbb{E}[(X - 1)^2] = \frac{20}{7}$$

1.4.7 Question 2 (Binomial Distribution)

In an AI-based quality control system, a manufacturer's memory chips are inspected automatically. Each chip has a 1% probability of being defective. A batch contains 20 chips.

If X is the number of defective chips in the batch, which of the following gives the correct value of:

- the probability that at most one chip is defective,
- the mean and variance of the number of non-defective chips?

Choices:

- A. $P(X \leq 1) = 0.98314$, $\mu = 19.8$, $\text{Var}(X) = 0.198$
B. $P(X \leq 1) = 0.81791$, $\mu = 0.2$, $\text{Var}(X) = 0.198$
C. $P(X \leq 1) = 0.98314$, $\mu = 20$, $\text{Var}(X) = 0.01$
D. $P(X \leq 1) = 0.90000$, $\mu = 19.8$, $\text{Var}(X) = 0.99$

Step 1: Identify Distribution

$$X \sim \text{Binomial}(n = 20, p = 0.01)$$

Step 2: Compute $P(X \leq 1)$

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

$$P(X = 0) = (0.99)^{20}$$

$$P(X = 1) = \binom{20}{1} (0.01)(0.99)^{19}$$

$$P(X \leq 1) \approx 0.98314$$

Step 3: Non-defective chips

Let Y = number of non-defective chips.

$$Y \sim \text{Binomial}(20, 0.99)$$

Step 4: Mean and Variance

$$\mu = np = 20(0.99) = 19.8$$

$$\text{Var}(Y) = np(1 - p) = 20(0.99)(0.01) = 0.198$$

Final Answer:

Option A

1.4.8 Question 3 (Continuous Distribution)

In an AI-based predictive maintenance application, the lifetime T (in months) of a light bulb is modeled by the probability density function:

$$f(t) = \begin{cases} k - \frac{t}{450}, & 0 \leq t \leq 30 \\ 0, & \text{otherwise} \end{cases}$$

Which of the following gives the correct values of the constant k and the probability that the bulb lasts at least 15 months?

Choices:

A. $k = \frac{1}{30}$, $P(T \geq 15) = \frac{1}{2}$

B. $k = \frac{1}{15}$, $P(T \geq 15) = \frac{1}{2}$

C. $k = \frac{2}{15}$, $P(T \geq 15) = \frac{1}{3}$

D. $k = \frac{1}{15}$, $P(T \geq 15) = \frac{1}{4}$

Step 1: Use total probability = 1

$$\int_0^{30} \left(k - \frac{t}{450} \right) dt = 1$$

$$\int_0^{30} k dt - \int_0^{30} \frac{t}{450} dt = 1$$

$$30k - \frac{1}{450} \cdot \frac{30^2}{2} = 1$$

$$30k - 1 = 1$$

$$30k = 2 \Rightarrow k = \frac{1}{15}$$

Step 2: Compute $P(T \geq 15)$

$$P(T \geq 15) = \int_{15}^{30} \left(\frac{1}{15} - \frac{t}{450} \right) dt$$

$$= \int_{15}^{30} \frac{1}{15} dt - \int_{15}^{30} \frac{t}{450} dt$$

$$= \left[\frac{t}{15} \right]_{15}^{30} - \left[\frac{t^2}{900} \right]_{15}^{30}$$

$$= (2 - 1) - \left(1 - \frac{1}{4} \right)$$

$$= 1 - \frac{3}{4} = \frac{1}{4}$$

Final Answer:

Option D

1.4.9 Question 4 (Normal Distribution — Reverse Z-Score Method)

In a data science model for anomaly-score analysis, the random variable X follows a normal distribution $N(\mu, \sigma)$.

Suppose you are given:

$$P(X < 66) = 0.0359, \quad P(X > 81) = 0.1151$$

Which of the following gives the correct values of μ , σ , and:

$$P(66 < X < 81)?$$

Choices:

A. $\mu = 70, \sigma = 6, P(66 < X < 81) = 0.7251$

B. $\mu = 75, \sigma = 3, P(66 < X < 81) = 0.8849$

C. $\mu = 72, \sigma = 5, P(66 < X < 81) = 0.9452$

D. $\mu = 75, \sigma = 5, P(66 < X < 81) = 0.7673$

Step 1: Convert probabilities to Z-values

From Z-tables:

$$P(Z < -1.8) = 0.0359 \Rightarrow Z = -1.8$$

$$P(Z > 1.2) = 0.1151 \Rightarrow Z = 1.2$$

Step 2: Build equations using $Z = \frac{X - \mu}{\sigma}$

For $X = 66$:

$$\frac{66 - \mu}{\sigma} = -1.8 \quad (1)$$

For $X = 81$:

$$\frac{81 - \mu}{\sigma} = 1.2 \quad (2)$$

Step 3: Solve the system

From (1):

$$66 - \mu = -1.8\sigma$$

From (2):

$$81 - \mu = 1.2\sigma$$

Subtract equations:

$$(81 - \mu) - (66 - \mu) = 1.2\sigma - (-1.8\sigma)$$

$$15 = 3\sigma$$

$$\sigma = 5$$

Now substitute into (1):

$$66 - \mu = -1.8 \times 5 = -9$$

$$\mu = 66 + 9 = 75$$

Step 4: Compute the required probability

$$P(66 < X < 81) = 1 - P(X < 66) - P(X > 81)$$

$$= 1 - 0.0359 - 0.1151 = 0.849$$

Step 5: Compare with choices

- The correct parameters are $\mu = 75$, $\sigma = 5$
- This matches Option D
- However, the probability in Option D is incorrect

Final Answer:

Option D has the correct parameters but the probability is incorrect.

$$P(66 < X < 81) = 0.849$$

1.4.10 Question from notes (Continuous PDF)

Find the value of c such that the function

$$f(x) = \begin{cases} 0, & x < 0 \\ cxe^{-x^2}, & x \geq 0 \end{cases}$$

is a valid probability density function (PDF).

Step 1: Use total probability = 1

$$\int_0^{\infty} cxe^{-x^2} dx = 1$$

$$c \int_0^{\infty} x e^{-x^2} dx = 1$$

Step 2: Rewrite the integral (trick from derivative)

Notice:

$$\frac{d}{dx}(e^{-x^2}) = -2xe^{-x^2}$$

So:

$$xe^{-x^2} = -\frac{1}{2} \cdot (-2xe^{-x^2})$$

$$\begin{aligned} \int xe^{-x^2} dx &= -\frac{1}{2} \int (-2xe^{-x^2}) dx \\ &= -\frac{1}{2} e^{-x^2} \end{aligned}$$

Step 3: Apply limits

$$\begin{aligned} \int_0^{\infty} xe^{-x^2} dx &= \left[-\frac{1}{2} e^{-x^2} \right]_0^{\infty} \\ &= -\frac{1}{2} (e^{-\infty} - e^0) \\ &= -\frac{1}{2} (0 - 1) = \frac{1}{2} \end{aligned}$$

Step 4: Solve for c

$$c \cdot \frac{1}{2} = 1$$

$$c = 2$$

Final Answer:

$$c = 2$$

1.4.11 Example from notes (Continuous Distribution — Time Headway)

Let the random variable X have the probability density function:

$$f(x) = \begin{cases} 0.15e^{-0.15(x-0.5)}, & x \geq 0.5 \\ 0, & \text{otherwise} \end{cases}$$

1. Show that $f(x)$ is a PDF
2. Find the mean $\mu = \mathbb{E}[X]$
3. Find the variance σ^2

4. Find $P(X \geq 2)$

—

Step 1: Verify PDF (total area = 1)

$$\int_{-\infty}^{\infty} f(x) dx = \int_{0.5}^{\infty} 0.15e^{-0.15(x-0.5)} dx$$

Let:

$$u = -0.15(x - 0.5)$$

$$\frac{d}{dx} (e^{-0.15(x-0.5)}) = -0.15e^{-0.15(x-0.5)}$$

So:

$$\int 0.15e^{-0.15(x-0.5)} dx = -e^{-0.15(x-0.5)}$$

Apply limits:

$$\left[-e^{-0.15(x-0.5)} \right]_{0.5}^{\infty}$$

$$= -(0 - 1) = 1$$

$\Rightarrow f(x)$ is a valid PDF

—

Step 2: Mean $\mu = \mathbb{E}[X]$

$$\mathbb{E}[X] = \int_{0.5}^{\infty} x \cdot 0.15e^{-0.15(x-0.5)} dx$$

Use integration by parts:

Let:

$$u = x \quad \Rightarrow \quad dv = dx$$

$$dv = 0.15e^{-0.15(x-0.5)} dx$$

$$v = -e^{-0.15(x-0.5)}$$

Then:

$$\int x \cdot 0.15e^{-0.15(x-0.5)} dx = -xe^{-0.15(x-0.5)} + \int e^{-0.15(x-0.5)} dx$$

Now:

$$\int e^{-0.15(x-0.5)} dx = \frac{-1}{0.15} e^{-0.15(x-0.5)}$$

So:

$$= -xe^{-0.15(x-0.5)} - \frac{1}{0.15} e^{-0.15(x-0.5)}$$

Apply limits:

$$\left[-xe^{-0.15(x-0.5)} - \frac{1}{0.15}e^{-0.15(x-0.5)} \right]_{0.5}^{\infty}$$

At $\infty \rightarrow 0$, at $x = 0.5$:

$$= 0 - \left[-0.5(1) - \frac{1}{0.15}(1) \right]$$

$$= 0.5 + \frac{1}{0.15}$$

$$= 0.5 + 6.\bar{6} = 7.1667$$

$$\mu \approx 7.17$$

Step 3: Variance σ^2

$$\mathbb{E}[X^2] = \int_{0.5}^{\infty} x^2 \cdot 0.15e^{-0.15(x-0.5)} dx$$

(Using same integration by parts method repeatedly)

Final result:

$$\mathbb{E}[X^2] = \frac{2}{(0.15)^2} + \frac{2(0.5)}{0.15} + (0.5)^2$$

$$= \frac{2}{0.0225} + \frac{1}{0.15} + 0.25$$

$$= 88.89 + 6.67 + 0.25 = 95.81$$

Now:

$$\sigma^2 = \mathbb{E}[X^2] - \mu^2$$

$$= 95.81 - (7.1667)^2$$

$$\approx 95.81 - 51.36 = 44.45$$

Step 4: Probability $P(X \geq 2)$

$$P(X \geq 2) = \int_2^{\infty} 0.15e^{-0.15(x-0.5)} dx$$

$$= \left[-e^{-0.15(x-0.5)} \right]_2^{\infty}$$

$$= 0 - \left(-e^{-0.15(2-0.5)}\right)$$

$$= e^{-0.15(1.5)} = e^{-0.225}$$

$$\approx 0.798$$

Final Answers:

$$\mu \approx 7.17, \quad \sigma^2 \approx 44.45, \quad P(X \geq 2) \approx 0.798$$

For more question $\Rightarrow$ solve sheet 6

1.5 Differentiation

1.5.1 Important Rules and Hints

Additional Resource:

Rules of Differentiation & Table of Differentiation & Implicit Differentiation (Google Drive)

1) Basic Derivatives

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x$$

—

2) Product Rule

$$(fg)' = f'g + fg'$$

—

3) Quotient Rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

—

4) Chain Rule (VERY IMPORTANT)

If:

$$y = f(g(x))$$

Then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

—

5) Higher Derivatives

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

—

6) Implicit Differentiation

Differentiate both sides and treat y as a function of x :

$$\frac{d}{dx}(y) = \frac{dy}{dx}$$

—

1.5.2 Gradient (Multivariable Calculus)

Definition:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

Important Idea:

Gradient points in direction of **maximum increase**

1.5.3 Chain Rule in Multiple Variables

If:

$$z = f(x, y), \quad x = g(t), \quad y = h(t)$$

Then:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

1.5.4 Lagrange Multipliers (Optimization)

Used when:

Max/Min with constraint

Rule:

$$\nabla f = \lambda \nabla g$$

System to solve:

$$\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x}$$

$$\frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y}$$

$$g(x, y) = c$$

1.5.5 Backpropagation (VERY IMPORTANT for AI)

Idea:

Apply chain rule through layers

Basic Rule:

$$\frac{dL}{dx} = \frac{dL}{dy} \cdot \frac{dy}{dx}$$

In Neural Networks:

- Forward: compute output
 - Backward: compute gradients
-

1.5.6 VERY IMPORTANT HINTS

- If function inside function → use **chain rule**
 - If multiplication → **product rule**
 - If division → **quotient rule**
 - In ML problems → always think **chain rule**
 - Gradient = vector of partial derivatives
 - Lagrange → always solve system, not one equation
 - Watch signs (very common mistakes)
 - Simplify before differentiating if possible
-

1.5.7 Mixed Differentiation Examples

Example 1

$$y = \pi^\pi + x^\pi + \pi^x$$

Solution:

$$\frac{d}{dx}(\pi^\pi) = 0 \quad (\text{constant})$$

$$\frac{d}{dx}(x^\pi) = \pi x^{\pi-1}$$

$$\frac{d}{dx}(\pi^x) = \pi^x \ln \pi$$

Final Answer:

$$\frac{dy}{dx} = \pi x^{\pi-1} + \pi^x \ln \pi$$

—

Example 2

$$y = \frac{\tan^3 x + \sqrt{x}}{\sec^4 x}$$

Solution (Quotient Rule):

Let:

$$f = \tan^3 x + \sqrt{x}, \quad g = \sec^4 x$$

$$f' = 3 \tan^2 x \sec^2 x + \frac{1}{2\sqrt{x}}$$

$$g' = 4 \sec^3 x (\sec x \tan x)$$

$$\frac{dy}{dx} = \frac{f'g - fg'}{g^2}$$

Final Answer (unsimplified):

$$\frac{dy}{dx} = \frac{(3 \tan^2 x \sec^2 x + \frac{1}{2\sqrt{x}}) \sec^4 x - (\tan^3 x + \sqrt{x}) \cdot 4 \sec^4 x \tan x}{\sec^8 x}$$

—

Example 3

$$y = x^{\sin x}$$

Solution (Logarithmic Differentiation):

$$\ln y = \sin x \ln x$$

Differentiate:

$$\frac{1}{y} \frac{dy}{dx} = \cos x \ln x + \frac{\sin x}{x}$$

$$\frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

—

Example 4 (Implicit)

$$x^{\sin y} = y^{\cos x}$$

Take $\ln$:

$$\sin y \ln x = \cos x \ln y$$

Differentiate:

$$\cos y \frac{dy}{dx} \ln x + \frac{\sin y}{x} = -\sin x \ln y + \frac{\cos x}{y} \frac{dy}{dx}$$

Group $\frac{dy}{dx}$:

$$\frac{dy}{dx} \left(\cos y \ln x - \frac{\cos x}{y} \right) = -\sin x \ln y - \frac{\sin y}{x}$$

Final Answer:

$$\frac{dy}{dx} = \frac{-\sin x \ln y - \frac{\sin y}{x}}{\cos y \ln x - \frac{\cos x}{y}}$$

—

Example 5

$$y = x^{\sin x} + \tan(x)^{\sqrt{x}}$$

Solution:

First term (from Example 3):

$$\frac{d}{dx}(x^{\sin x}) = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

Second term:

Let:

$$u = \tan(x)^{\sqrt{x}}$$

$$\ln u = \sqrt{x} \ln(\tan x)$$

Differentiate:

$$\frac{1}{u} \frac{du}{dx} = \frac{1}{2\sqrt{x}} \ln(\tan x) + \sqrt{x} \cdot \frac{\sec^2 x}{\tan x}$$

$$\frac{du}{dx} = \tan(x)^{\sqrt{x}} \left(\frac{\ln(\tan x)}{2\sqrt{x}} + \sqrt{x} \frac{\sec^2 x}{\tan x} \right)$$

Final Answer:

$$\frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right) + \tan(x)^{\sqrt{x}} \left(\frac{\ln(\tan x)}{2\sqrt{x}} + \sqrt{x} \frac{\sec^2 x}{\tan x} \right)$$

—

Example 6

$$y = \sinh^{-1}(x) + \cosh(e^{3x})^4 + \tan^{-1}(\ln^3 x)$$

Solution:

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}[\cosh(e^{3x})^4] = 4 \cosh^3(e^{3x}) \cdot \sinh(e^{3x}) \cdot 3e^{3x}$$

$$\frac{d}{dx}[\tan^{-1}(\ln^3 x)] = \frac{1}{1+\ln^6 x} \cdot \frac{3 \ln^2 x}{x}$$

Final Answer:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}} + 12e^{3x} \cosh^3(e^{3x}) \sinh(e^{3x}) + \frac{3 \ln^2 x}{x(1+\ln^6 x)}$$

—

Example 7

$$y = \tan(x)^{\cos(x^2)}$$

Solution (Logarithmic Differentiation):

$$\ln y = \cos(x^2) \ln(\tan x)$$

Differentiate:

$$\frac{1}{y} \frac{dy}{dx} = -2x \sin(x^2) \ln(\tan x) + \cos(x^2) \frac{\sec^2 x}{\tan x}$$

$$\frac{dy}{dx} = \tan(x)^{\cos(x^2)} \left[-2x \sin(x^2) \ln(\tan x) + \cos(x^2) \frac{\sec^2 x}{\tan x} \right]$$

—

1.5.8 Chain Rule Examples

Example 1 (Chain Rule with Single Variable Parameter)

Given:

$$z = f(x, y) = x^2 e^y, \quad x(t) = t^2 - 1, \quad y(t) = \sin t$$

Find:

$$\frac{d}{dt} f(x(t), y(t))$$

Solution:

Step 1: Compute partial derivatives

$$\frac{\partial z}{\partial x} = 2x e^y, \quad \frac{\partial z}{\partial y} = x^2 e^y$$

Step 2: Compute derivatives of inner functions

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = \cos t$$

Step 3: Apply Chain Rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Step 4: Substitute

$$= (2xe^y)(2t) + (x^2e^y)(\cos t)$$

Now substitute $x = t^2 - 1$, $y = \sin t$:

$$= 2(t^2 - 1)e^{\sin t}(2t) + (t^2 - 1)^2e^{\sin t} \cos t$$

Final Answer:

$4t(t^2 - 1)e^{\sin t} + (t^2 - 1)^2e^{\sin t} \cos t$

—

Example 2 (Chain Rule with Two Variables)

Suppose:

$$f(x, y) = e^{xy}, \quad x(u, v) = 3u \sin v, \quad y(u, v) = 4v^2u$$

Let:

$$g(u, v) = f(x(u, v), y(u, v))$$

Find:

$$\frac{\partial g}{\partial u}, \quad \frac{\partial g}{\partial v}$$

Solution:

Step 1: Compute partial derivatives of outer function

$$\frac{\partial f}{\partial x} = ye^{xy}, \quad \frac{\partial f}{\partial y} = xe^{xy}$$

—

Step 2: Compute partial derivatives of inner functions

$$\frac{\partial x}{\partial u} = 3 \sin v, \quad \frac{\partial x}{\partial v} = 3u \cos v$$

$$\frac{\partial y}{\partial u} = 4v^2, \quad \frac{\partial y}{\partial v} = 8vu$$

—

Step 3: Apply Chain Rule

$$\begin{aligned} \frac{\partial g}{\partial u} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \\ &= (ye^{xy})(3 \sin v) + (xe^{xy})(4v^2) \end{aligned}$$

—

$$\begin{aligned}\frac{\partial g}{\partial v} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} \\ &= (ye^{xy})(3u \cos v) + (xe^{xy})(8vu)\end{aligned}$$

—

Step 4 (Optional Simplification)

You can factor:

$$e^{xy}$$

$$\begin{aligned}\frac{\partial g}{\partial u} &= e^{xy}(3y \sin v + 4v^2x) \\ \frac{\partial g}{\partial v} &= e^{xy}(3uy \cos v + 8vux)\end{aligned}$$

Final Answer:

$$\boxed{\frac{\partial g}{\partial u} = (ye^{xy})(3 \sin v) + (xe^{xy})(4v^2)}$$

$$\boxed{\frac{\partial g}{\partial v} = (ye^{xy})(3u \cos v) + (xe^{xy})(8vu)}$$

1.5.9 Gradient Examples

Important Rules and Hints (VERY IMPORTANT)

1) Gradient of a Function

If

$$f(x, y)$$

then the gradient is:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

For three variables:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

—

2) Meaning of Gradient

- The gradient points in the direction of **maximum increase**.
- The negative gradient points in the direction of **maximum decrease**.
- The magnitude of the gradient gives the **maximum rate of change**.

3) Maximum and Minimum Rates of Change

$$\text{Maximum rate of change} = \|\nabla f\|$$

$$\text{Minimum rate of change} = -\|\nabla f\|$$

4) Direction of Maximum Increase

$$\mathbf{u} = \frac{\nabla f}{\|\nabla f\|}$$

5) Gradient Descent Rule

To minimize a function:

$$x_{\text{new}} = x_{\text{old}} - \alpha f'(x_{\text{old}})$$

In more variables:

$$\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla L(\theta_{\text{old}})$$

where α is the learning rate.

6) Hints

- If the question asks for **maximum rate**, compute $\|\nabla f\|$.
- If the question asks for **minimum rate**, use $-\|\nabla f\|$.
- If the question is about optimization, think of the **negative gradient**.
- Always check whether the problem is one-variable or multivariable.
- In machine learning, gradient descent means moving **downhill**.

Example 1 (Maximum and Minimum Rates of Change)

Find the maximum and minimum rates of change of:

$$f(x, y) = x^2 + y^2$$

at the point $(1, 3)$.

Solution:

Step 1: Compute the gradient

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x, 2y)$$

At $(1, 3)$:

$$\nabla f(1, 3) = (2, 6)$$

Step 2: Find the magnitude of the gradient

$$\|\nabla f(1, 3)\| = \sqrt{2^2 + 6^2} = \sqrt{40}$$

Step 3: Maximum rate of change

$$\text{Maximum rate} = \|\nabla f(1, 3)\| = \sqrt{40}$$

Direction of maximum increase:

$$\mathbf{u} = \frac{\nabla f(1, 3)}{\|\nabla f(1, 3)\|} = \frac{(2, 6)}{\sqrt{40}}$$

Step 4: Minimum rate of change

$$\text{Minimum rate} = -\|\nabla f(1, 3)\| = -\sqrt{40}$$

Direction of minimum decrease:

$$\mathbf{u} = -\frac{(2, 6)}{\sqrt{40}}$$

Final Answer:

$$\boxed{\text{Maximum rate} = \sqrt{40}}, \quad \boxed{\text{Minimum rate} = -\sqrt{40}}$$

—

Example 2 (One Step of Gradient Descent)

Let:

$$f(x) = x^2 + 5x + 10$$

Find the next point if:

$$x_0 = 2, \quad \alpha = 0.1$$

Solution:

Step 1: Find the derivative

$$f'(x) = 2x + 5$$

Step 2: Evaluate the derivative at $x_0 = 2$

$$f'(2) = 2(2) + 5 = 9$$

Step 3: Use the gradient descent update

$$x_1 = x_0 - \alpha f'(x_0)$$

$$x_1 = 2 - 0.1(9) = 2 - 0.9 = 1.1$$

Step 4: Check the function value

$$f(2) = 2^2 + 5(2) + 10 = 24$$

$$f(1.1) = (1.1)^2 + 5(1.1) + 10 = 1.21 + 5.5 + 10 = 16.71$$

Final Answer:

$$\boxed{x_1 = 1.1}$$

1.5.10 Lagrange Multiplier Examples

Important Rules and Hints

1) Main Rule

To optimize a function $f(x, y)$ subject to a constraint $g(x, y) = 0$:

$$\nabla f = \lambda \nabla g$$

—

2) System to Solve

$$\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x}$$

$$\frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y}$$

$$g(x, y) = 0$$

—

3) Hint

If the question says:

- maximize volume,
- minimize distance,
- optimize with a condition,

then use **Lagrange multipliers**.

—

4) Important Idea

At the optimum:

$$\nabla f \text{ is parallel to } \nabla g$$

—

Example 1 (Maximizing the Volume of a Cylinder)

A closed cylindrical tin with top and bottom is made from sheet metal. The total surface area is:

$$A_0 = 24\pi$$

Let the cylinder have radius r and height h .

Find the dimensions that maximize the volume.

Solution:

Step 1: Write the volume function

$$V(r, h) = \pi r^2 h$$

Step 2: Write the constraint

$$2\pi r^2 + 2\pi r h = 24\pi$$

So:

$$g(r, h) = 2\pi r^2 + 2\pi r h - 24\pi = 0$$

Step 3: Form the Lagrangian

$$\mathcal{L}(r, h, \lambda) = \pi r^2 h + \lambda(2\pi r^2 + 2\pi r h - 24\pi)$$

Step 4: Compute partial derivatives

$$\frac{\partial \mathcal{L}}{\partial r} = 2\pi r h + \lambda(4\pi r + 2\pi h) = 0$$

$$\frac{\partial \mathcal{L}}{\partial h} = \pi r^2 + \lambda(2\pi r) = 0$$

$$2\pi r^2 + 2\pi r h - 24\pi = 0$$

Step 5: Simplify the second equation

$$\pi r^2 + 2\pi \lambda r = 0$$

Divide by πr :

$$r + 2\lambda = 0$$

$$\lambda = -\frac{r}{2}$$

Step 6: Substitute into the first equation

$$2\pi r h - \frac{r}{2}(4\pi r + 2\pi h) = 0$$

$$2\pi r h - 2\pi r^2 - \pi r h = 0$$

$$\pi r h = 2\pi r^2$$

$$h = 2r$$

Step 7: Use the constraint

$$2\pi r^2 + 2\pi r(2r) = 24\pi$$

$$2\pi r^2 + 4\pi r^2 = 24\pi$$

$$6\pi r^2 = 24\pi$$

$$r^2 = 4 \Rightarrow r = 2$$

Then:

$$h = 2r = 4$$

Step 8: Compute the maximum volume

$$V_{\max} = \pi r^2 h = \pi(2^2)(4) = 16\pi$$

Final Answer:

$$\boxed{r = 2, \quad h = 4, \quad V_{\max} = 16\pi}$$

—

Example 2 (Closest Point on a Line to the Origin)

Use Lagrange multipliers to find the point on the line:

$$y = 3 - 2x$$

that is closest to the origin.

Solution:

Step 1: Minimize the squared distance

$$f(x, y) = x^2 + y^2$$

Step 2: Write the constraint

$$y = 3 - 2x \Rightarrow 2x + y - 3 = 0$$

So:

$$g(x, y) = 2x + y - 3 = 0$$

Step 3: Compute gradients

$$\nabla f = (2x, 2y), \quad \nabla g = (2, 1)$$

Step 4: Apply Lagrange condition

$$\nabla f = \lambda \nabla g$$

So:

$$(2x, 2y) = \lambda(2, 1)$$

This gives:

$$2x = 2\lambda \Rightarrow x = \lambda$$

$$2y = \lambda$$

From these:

$$x = 2y$$

Step 5: Use the constraint

$$2x + y - 3 = 0$$

Substitute $x = 2y$:

$$2(2y) + y - 3 = 0$$

$$5y = 3$$

$$y = \frac{3}{5}$$

Then:

$$x = 2y = \frac{6}{5}$$

Step 6: Find the minimum distance

The minimum squared distance is:

$$f_{\min} = x^2 + y^2$$

$$f_{\min} = \left(\frac{6}{5}\right)^2 + \left(\frac{3}{5}\right)^2 = \frac{36}{25} + \frac{9}{25} = \frac{45}{25} = \frac{9}{5}$$

So the minimum distance is:

$$d_{\min} = \sqrt{\frac{9}{5}} = \frac{3}{\sqrt{5}}$$

Final Answer:

$$\left(\frac{6}{5}, \frac{3}{5}\right)$$

$$d_{\min} = \frac{3}{\sqrt{5}}$$

1.5.11 Backpropagation Examples

Important Rules and Hints (VERY IMPORTANT)

1) Main Idea

Backpropagation is just the **chain rule** applied backward through a computational graph.

—

2) Basic Formula

If a loss L depends on a variable x through intermediate steps, then:

$$\frac{dL}{dx} = \frac{dL}{dy} \cdot \frac{dy}{dx}$$

—

3) In Neural Networks

- Forward pass: compute the output

- Backward pass: compute gradients
- Update weights using gradient descent

4) Common Derivative of Sigmoid

If:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

then:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

5) Hint

When you see a network or a composition of functions:

- first write the forward pass,
- then differentiate step by step,
- then multiply the local derivatives.

Example 1 (Backpropagation Through One Neuron)

Let:

$$\begin{aligned} z &= wx + b \\ a &= \sigma(z) = \frac{1}{1 + e^{-z}} \\ L &= (a - y)^2 \end{aligned}$$

Find:

$$\frac{dL}{dw}$$

Solution:

Step 1: Differentiate the loss with respect to a

$$\frac{dL}{da} = 2(a - y)$$

Step 2: Differentiate a with respect to z

$$\frac{da}{dz} = a(1 - a)$$

Step 3: Differentiate z with respect to w

$$\frac{dz}{dw} = x$$

Step 4: Apply the chain rule

$$\frac{dL}{dw} = \frac{dL}{da} \cdot \frac{da}{dz} \cdot \frac{dz}{dw}$$

$$\frac{dL}{dw} = 2(a - y) \cdot a(1 - a) \cdot x$$

Final Answer:

$$\boxed{\frac{dL}{dw} = 2(a - y)a(1 - a)x}$$

Example 2 (Backpropagation Idea with a Simple Computation Graph)

Suppose:

$$u = x^2$$

$$v = 3u$$

$$L = v^2$$

Find:

$$\frac{dL}{dx}$$

Solution:

Step 1: Differentiate each layer

$$\frac{dL}{dv} = 2v$$

$$\frac{dv}{du} = 3$$

$$\frac{du}{dx} = 2x$$

Step 2: Multiply derivatives backward

$$\frac{dL}{dx} = \frac{dL}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx}$$

$$= (2v)(3)(2x)$$

Step 3: Substitute $v = 3u = 3x^2$

$$\frac{dL}{dx} = 12vx = 12(3x^2)x = 36x^3$$

Final Answer:

$$\boxed{\frac{dL}{dx} = 36x^3}$$

1.5.12 General Problem from extra Question pdf

Consider the quadratic function:

$$f(x) = 3x^2 + 6x + 5$$

Answer the following:

- (a) Compute the derivative $f'(x)$.
- (b) Determine whether the function is convex. Justify your answer.
- (c) Perform two iterations of gradient descent starting from the initial point:

$$x_0 = 2$$

using a learning rate:

$$\alpha = 0.1$$

- (d) Find the exact minimizer x^* by solving:

$$f'(x) = 0$$

and compute the minimum value $f(x^*)$.

- (e) Compare the point obtained after two iterations with the exact minimizer by computing:

$$|x_2 - x^*|$$

For a quadratic function, the error in gradient descent reduces by a factor:

$$|1 - \alpha f''|$$

per iteration. Use this to estimate how many additional iterations are needed to reach an error less than 0.001.

- (f) Suppose the learning rate is increased to:

$$\alpha = 0.4$$

Perform only the first iteration starting from $x_0 = 2$, compute the new point, and evaluate the function at that point.

Compare it with the initial value and determine whether this learning rate leads to convergence. Justify your answer.

(a) Compute the derivative

Step 1: Differentiate

$$f'(x) = 6x + 6$$

$$f'(x) = 6x + 6$$

(b) Check convexity

Step 1: Compute second derivative

$$f''(x) = 6$$

Step 2: Conclusion

Since:

$$f''(x) > 0$$

The function is convex

(c) Two iterations of Gradient Descent

Given:

$$x_0 = 2, \quad \alpha = 0.1$$

Iteration 1:

$$f'(2) = 6(2) + 6 = 18$$

$$x_1 = x_0 - \alpha f'(x_0) = 2 - 0.1(18) = 2 - 1.8 = 0.2$$

Iteration 2:

$$f'(0.2) = 6(0.2) + 6 = 7.2$$

$$x_2 = 0.2 - 0.1(7.2) = 0.2 - 0.72 = -0.52$$

$x_2 = -0.52$

(d) Exact minimizer

Step 1: Solve $f'(x) = 0$

$$6x + 6 = 0$$

$$x^* = -1$$

Step 2: Compute minimum value

$$f(-1) = 3(1) - 6 + 5 = 2$$

$$\boxed{x^* = -1, \quad f(x^*) = 2}$$

(e) Error after two iterations

$$|x_2 - x^*| = |-0.52 - (-1)| = |0.48| = 0.48$$

$$\boxed{\text{Error} = 0.48}$$

Error Reduction Factor

$$|1 - \alpha f''| = |1 - 0.1(6)| = 0.4$$

So error reduces by factor 0.4 each step.

Estimate number of iterations

We want:

$$(0.4)^k \cdot 0.48 < 0.001$$

Approximation:

$$(0.4)^5 \approx 0.01, \quad (0.4)^7 \approx 0.0016$$

So:

$$k \approx 7 \text{ more iterations}$$

$$\boxed{\text{About 7 more iterations needed}}$$

(f) Larger learning rate $\alpha = 0.4$

Step 1: First iteration

$$x_1 = 2 - 0.4(18) = 2 - 7.2 = -5.2$$

Step 2: Compare function values

$$f(2) = 29$$

$$f(-5.2) = 3(27.04) - 31.2 + 5 = 81.12 - 31.2 + 5 = 54.92$$

Conclusion

$$f(-5.2) > f(2)$$

The function increased $\rightarrow$ divergence

Learning rate $\alpha = 0.4$ is too large

1.6 Limits & L'Hôpital Rule

1.6.1 Important Rules and Hints (VERY IMPORTANT)

1) L'Hôpital Rule

If:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

Then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

—

2) Conditions to Apply Rule

- Must be $\frac{0}{0}$ or $\frac{\infty}{\infty}$
- Functions must be differentiable

—

3) Important Indeterminate Forms

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 1^\infty, \quad 0^0, \quad \infty^0$$

—

4) (VERY IMPORTANT)

- $0 \cdot \infty \rightarrow$ convert to fraction
- $\infty - \infty \rightarrow$ combine into one fraction
- $a^b \rightarrow$ take $\ln$

—

Exam Hints

- Always check the form FIRST
- You can apply L'Hôpital more than once
- If not indeterminate $\rightarrow$ DO NOT use it
- In powers $\rightarrow$ take $\ln$ first

—

1.6.2 Case 1: $\frac{0}{0}$ Form

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Step 1: Check form

$$\frac{0}{0}$$

Step 2: Apply L'Hôpital

$$= \lim_{x \rightarrow 0} \frac{\cos x}{1}$$

$$= \cos(0) = 1$$

Final Answer:

$$1$$

—

1.6.3 Case 2: $\frac{\infty}{\infty}$ Form

$$\lim_{x \rightarrow \infty} \frac{x}{e^x}$$

Step 1: Check form

$$\frac{\infty}{\infty}$$

Step 2: Apply L'Hôpital

$$= \lim_{x \rightarrow \infty} \frac{1}{e^x}$$

$$= 0$$

Final Answer:

$$0$$

—

1.6.4 Case 3: $0 \cdot \infty$ Form

$$\lim_{x \rightarrow 0^+} x \ln x$$

Step 1: Identify form

$$0 \cdot (-\infty)$$

Step 2: Convert to fraction

$$\begin{aligned} &= \frac{\ln x}{1/x} \\ &= \frac{-\infty}{\infty} \end{aligned}$$

Step 3: Apply L'Hôpital

$$\begin{aligned} &= \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} \\ &= \lim_{x \rightarrow 0^+} (-x) \\ &= 0 \end{aligned}$$

Final Answer:

$$0$$

—

1.6.5 Case 4: 0^0 Form (Power with Trigonometric Functions)

$$\lim_{x \rightarrow 0} (\tan x)^{\sin x}$$

Step 1: Identify the form

As $x \rightarrow 0$:

$$\tan x \rightarrow 0, \quad \sin x \rightarrow 0$$

$$\Rightarrow 0^0 \text{ (Indeterminate)}$$

—

Step 2: Let

$$y = (\tan x)^{\sin x}$$

Take $\ln$:

$$\ln y = \sin x \cdot \ln(\tan x)$$

—

Step 3: Convert to fraction (to use L'Hôpital)

$$\ln y = \frac{\ln(\tan x)}{1/\sin x}$$

As $x \rightarrow 0$:

$$\frac{-\infty}{\infty} \text{ (Indeterminate)}$$

Step 4: Apply L'Hôpital

Differentiate numerator and denominator:

$$\frac{d}{dx}[\ln(\tan x)] = \frac{1}{\tan x} \cdot \sec^2 x$$

$$\frac{d}{dx}[1/\sin x] = -\frac{\cos x}{\sin^2 x}$$

So:

$$\ln y = \lim_{x \rightarrow 0} \frac{\frac{\sec^2 x}{\tan x}}{-\frac{\cos x}{\sin^2 x}}$$

Step 5: Simplify

$$\begin{aligned} &= \lim_{x \rightarrow 0} \left(\frac{\sec^2 x}{\tan x} \cdot \frac{\sin^2 x}{-\cos x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{1}{\cos^2 x} \cdot \frac{\sin^2 x}{\tan x \cdot (-\cos x)} \right) \end{aligned}$$

Using:

$$\begin{aligned} \tan x &= \frac{\sin x}{\cos x} \\ &= \lim_{x \rightarrow 0} \left(\frac{1}{\cos^2 x} \cdot \frac{\sin^2 x}{\frac{\sin x}{\cos x} \cdot (-\cos x)} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{1}{\cos^2 x} \cdot \frac{\sin^2 x}{-\sin x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{-\sin x}{\cos^2 x} \right) \\ &= 0 \end{aligned}$$

Step 6: Return to y

$$\ln y = 0 \Rightarrow y = e^0 = 1$$

Final Answer:

$$\boxed{1}$$

1.6.6 Case 5: 1^∞ Form (Important)

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

Step 1: Let

$$y = \left(1 + \frac{1}{x}\right)^x$$

Step 2: Take $\ln$

$$\ln y = x \ln \left(1 + \frac{1}{x}\right)$$

Step 3: Convert to fraction

$$= \frac{\ln(1 + 1/x)}{1/x}$$

Step 4: Apply L'Hôpital

$$= \lim_{x \rightarrow \infty} \frac{\frac{-1}{x^2(1+1/x)}}{-1/x^2}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 + 1/x}$$

$$= 1$$

Step 5: Return to y

$$\ln y = 1 \Rightarrow y = e$$

Final Answer:

$$e$$

—

1.6.7 Extra Important Forms

- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$

Additional Resource:

[Full L'Hôpital Notes \(Google Drive\)](#)

1.7 Metrics may come in exam

1.7.1 Regression Metrics

1) MAE — Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Meaning: Average absolute difference between actual and predicted values.

Advantage: Easy to understand.

2) MSE — Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Meaning: Average squared error.

Advantage: Penalizes large errors more strongly.

3) RMSE — Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

Meaning: Same as MSE but in original units.

4) R^2 Score (Coefficient of Determination)

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Meaning: Measures how much variance is explained by the model.

- $R^2 = 1 \rightarrow$ Perfect fit
 - $R^2 = 0 \rightarrow$ Same as predicting mean
 - $R^2 < 0 \rightarrow$ Worse than mean prediction
-

1.7.2 Classification Metrics

Confusion Matrix Terms

- TP = True Positive
- TN = True Negative
- FP = False Positive

- FN = False Negative

—

1) Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Meaning: Overall correctness.

—

2) Precision

$$Precision = \frac{TP}{TP + FP}$$

Meaning: How many predicted positives are actually correct.

—

3) Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

Meaning: How many actual positives are detected.

—

4) Specificity

$$Specificity = \frac{TN}{TN + FP}$$

Meaning: How many actual negatives are correctly identified.

—

5) F1-Score

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Meaning: Balance between Precision and Recall.

—

1.7.3 Important Exam Notes

- Use MAE when you want simple interpretation
- Use MSE/RMSE when large errors matter
- Use R^2 to evaluate overall fit
- Accuracy is not reliable for imbalanced data
- Precision important when FP is costly
- Recall important when FN is costly
- F1-score balances Precision and Recall

2 Written Part (Gamma Function)

2.1 Important Summary Rules (VERY IMPORTANT)

1) Definition:

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

2) Recursion Rule:

$$\Gamma(n+1) = n\Gamma(n)$$

3) Factorial Relation:

$$\Gamma(n+1) = n!$$

4) Special Value:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

5) General Formula:

$$\int_0^{\infty} x^n e^{-x} dx = \Gamma(n+1)$$

6) Important Substitution: If you see $e^{-x^2} \rightarrow$ use:

$$x^2 = t$$

7) Logarithm Integral Rule:

$$\int_0^1 x^m (\ln x)^n dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$$

2.2 Important Hints

- Always try to match Gamma form.
- If power is not integer $\rightarrow$ Gamma still works.
- If e^{-x^2} appears $\rightarrow$ substitution is required.
- Log integrals usually have direct formula.

2.3 Solved Examples

2.3.1 Example 1

$$\int_0^{\infty} x^3 e^{-x} dx$$

Solution:

Compare with:

$$\int_0^{\infty} x^n e^{-x} dx = \Gamma(n+1)$$

Here $n = 3$

$$= \Gamma(4)$$

Using factorial rule:

$$\Gamma(4) = 3! = 6$$

Final Answer:

$$6$$

2.3.2 Example 2

$$\int_0^{\infty} x^{5.2} e^{-x^2} dx$$

Step 1: Substitution

Let:

$$t = x^2 \Rightarrow dt = 2x dx$$

$$x dx = \frac{dt}{2}$$

Rewrite:

$$x^{5.2} = (x^2)^{2.6} = t^{2.6}$$

Integral becomes:

$$\begin{aligned} & \int_0^{\infty} t^{2.6} e^{-t} \cdot \frac{dt}{2\sqrt{t}} \\ &= \frac{1}{2} \int_0^{\infty} t^{2.6-0.5} e^{-t} dt \\ &= \frac{1}{2} \int_0^{\infty} t^{2.1} e^{-t} dt \\ &= \frac{1}{2} \Gamma(3.1) \end{aligned}$$

Final Answer:

$$\frac{1}{2} \Gamma(3.1)$$

2.3.3 Example 3

$$\int_0^1 \frac{1}{\sqrt{-\ln x}} dx$$

Substitution:

Let:

$$t = -\ln x$$

Then:

$$dx = e^{-t} dt$$

Integral becomes:

$$\begin{aligned} \int_0^\infty \frac{e^{-t}}{\sqrt{t}} dt \\ = \Gamma\left(\frac{1}{2}\right) \\ = \sqrt{\pi} \end{aligned}$$

Final Answer:

$$\sqrt{\pi}$$

2.3.4 Example 4

$$\int_0^3 (\ln(3/x))^{3/2} dx$$

Substitution:

Let:

$$t = \ln(3/x)$$

Then:

$$x = 3e^{-t}, \quad dx = -3e^{-t} dt$$

Change limits:

When $x = 3 \Rightarrow t = 0$

When $x \rightarrow 0 \Rightarrow t \rightarrow \infty$

Integral becomes:

$$\begin{aligned} \int_0^\infty t^{3/2} \cdot 3e^{-t} dt \\ = 3 \int_0^\infty t^{3/2} e^{-t} dt \\ = 3\Gamma\left(\frac{5}{2}\right) \end{aligned}$$

$$= 3 \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}$$

$$= \frac{9}{4} \sqrt{\pi}$$

Final Answer:

$$\frac{9}{4} \sqrt{\pi}$$

2.3.5 Example 5 (Derivation + Application)

$$I = \int_0^1 x^m (\ln x)^n dx$$

Step 1: Substitution

Let:

$$t = -\ln x \Rightarrow \ln x = -t$$

Differentiate:

$$dx = e^{-t} dt$$

Also:

$$x = e^{-t} \Rightarrow x^m = e^{-mt}$$

Change limits:

When $x = 0 \Rightarrow t = \infty$

When $x = 1 \Rightarrow t = 0$

Step 2: Substitute into integral

$$I = \int_{\infty}^0 e^{-mt} (-t)^n e^{-t} dt$$

$$= \int_{\infty}^0 (-1)^n t^n e^{-(m+1)t} dt$$

Reverse limits:

$$= (-1)^n \int_0^{\infty} t^n e^{-(m+1)t} dt$$

Step 3: Convert to Gamma form

We know:

$$\int_0^{\infty} t^n e^{-at} dt = \frac{\Gamma(n+1)}{a^{n+1}}$$

Here $a = m+1$

$$I = (-1)^n \frac{\Gamma(n+1)}{(m+1)^{n+1}}$$

Step 4: Use factorial relation

$$\Gamma(n+1) = n!$$

$$\boxed{\int_0^1 x^m (\ln x)^n dx = \frac{(-1)^n n!}{(m+1)^{n+1}}}$$

—
Now Apply the Rule

$$m = 7, \quad n = 9$$

$$\begin{aligned} I &= \frac{(-1)^9 9!}{(7+1)^{10}} \\ &= \frac{-9!}{8^{10}} \end{aligned}$$

Final Answer:

$$\frac{-9!}{8^{10}}$$

2.3.6 Extra Questions

1. $\int_0^\infty x^2 e^{-x} dx = \Gamma(3) = 2! = 2$
2. $\int_0^\infty x^{1/2} e^{-x} dx = \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}$
3. $\int_0^\infty e^{-ax} x^n dx = \frac{\Gamma(n+1)}{a^{n+1}}$
4. $\int_0^1 (\ln x)^3 dx = -6$
5. $\int_0^\infty x^4 e^{-2x} dx = \frac{\Gamma(5)}{2^5} = \frac{4!}{32} = \frac{24}{32} = \frac{3}{4}$

2.4 General Answer Forms

$$\int_0^\infty x^n e^{-ax} dx = \frac{\Gamma(n+1)}{a^{n+1}}$$

2.5 Final Tips Before Exam

- Memorize the identity: $\Gamma(n+1) = n!$
- Any integral involving $e^{-x} \rightarrow$ think Gamma function
- Any integral involving $e^{-x^2} \rightarrow$ use substitution
- Logarithmic integrals $\rightarrow$ apply the known formula directly
- Review carefully: $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

3 Quiz 1

Question 1

Let A and B be 3×3 matrices. Suppose that

$$(A^T - 3I)^{-1} = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, \quad \text{and} \quad |2B| = 8,$$

where I is the 3×3 identity matrix and $|\cdot|$ denotes the determinant.

(1) Find the matrix A :

A. $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

B. $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

C. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

D. $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Solution:

$$(A^T - 3I)^{-1} = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

Taking inverse:

$$A^T - 3I = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Add $3I$:

$$A^T = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Since the matrix is diagonal:

$$A = A^T = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Correct answer: B

(2) Find $\det(B^{-1})$:

- A. 8
- B. 2
- C. 1
- D. $\frac{1}{8}$

Solution:

$$|2B| = 2^3|B| = 8|B|$$

$$8|B| = 8 \Rightarrow |B| = 1$$

$$|B^{-1}| = \frac{1}{|B|} = 1$$

Correct answer: C

(3) Find $|2AB|$:

- A. 1920
- B. 1915
- C. 1905
- D. 1900

Solution:

$$|2AB| = 2^3|A||B| = 8|A||B|$$

$$|B| = 1$$

$$|A| = 8 \times 6 \times 5 = 240$$

$$|2AB| = 8 \times 240 \times 1 = 1920$$

Correct answer: A

Q2: Decode the Message

Question:

Use the given table to decode the message:

20, 1, 27, 5, 8, 8, 22, 13, 14, 14, 38, 23

using the matrix:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Letter Mapping: $A = 1, B = 2, \dots, Z = 26$, and $0 = \text{space}$.

Choices:

- A. SAVE HIM NEW
- B. SAVE HER NOW
- C. SAVE HIM NOW
- D. SAVE HIM NOT

Solution:

Step 1: Group the numbers into pairs:

(20, 1), (27, 5), (8, 8), (22, 13), (14, 14), (38, 23)

Step 2: Find the inverse of matrix A :

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Step 3: Decode each pair by multiplying:

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a - b \\ b \end{bmatrix}$$

Apply to each pair:

$$(20, 1) \rightarrow (20 - 1, 1) = (19, 1)$$

$$(27, 5) \rightarrow (22, 5)$$

$$(8, 8) \rightarrow (0, 8)$$

$$(22, 13) \rightarrow (9, 13)$$

$$(14, 14) \rightarrow (0, 14)$$

$$(38, 23) \rightarrow (15, 23)$$

Step 4: Convert numbers to letters:

$$19 = S, 1 = A, 22 = V, 5 = E, 0 = \text{space}, 8 = H$$

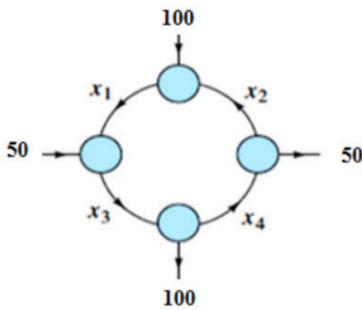
$$9 = I, 13 = M, 0 = \text{space}, 15 = O, 23 = W$$

Step 5: Combine:

SAVE HIM NOW

Final Answer: C. SAVE HIM NOW

Q3(A): Flow Conservation Model



Question: Which linear system correctly models the loop using flow conservation?

Choices:

A.

$$\begin{cases} 100 + x_1 = x_2 \\ 50 + x_3 = x_1 \\ x_4 = 100 + x_3 \\ x_2 = 50 + x_4 \end{cases}$$

B.

$$\begin{cases} 100 + x_2 = x_1 \\ 50 + x_1 = x_3 \\ x_3 = 50 + x_4 \\ x_4 = 100 + x_2 \end{cases}$$

C.

$$\begin{cases} 100 + x_2 = x_1 \\ 50 + x_1 = x_3 \\ x_3 = 100 + x_4 \\ x_4 = 50 + x_2 \end{cases}$$

D.

$$\begin{cases} 100 + x_2 = x_3 \\ 50 + x_1 = x_2 \\ x_3 = 100 + x_4 \\ x_4 = 50 + x_1 \end{cases}$$

Solution: We apply conservation of flow (Input = Output) at each junction.

North junction:

$$\text{Inflow: } 100 + x_2, \quad \text{Outflow: } x_1 \Rightarrow 100 + x_2 = x_1$$

West junction:

$$\text{Inflow: } 50 + x_1, \quad \text{Outflow: } x_3 \Rightarrow 50 + x_1 = x_3$$

South junction:

$$\text{Inflow: } x_3, \quad \text{Outflow: } 100 + x_4 \Rightarrow x_3 = 100 + x_4$$

East junction:

$$\text{Inflow: } x_4, \quad \text{Outflow: } 50 + x_2 \Rightarrow x_4 = 50 + x_2$$

This matches option C.

Final Answer: C

Q3(B): Interpretation of Infinite Solutions

Question: From an AI/data-science viewpoint, what does it mean if the system has infinitely many solutions?

Choices:

- A. The measurements constrain the flows, but one internal loop flow is still a free parameter (multiple valid states fit the same boundary data).

- B. The sensors are wrong and must be replaced.
- C. The loop must be closed permanently.
- D. The network is unstable and cannot be controlled.

Solution:

Step 1: A system with infinitely many solutions means it is **underdetermined**.

Step 2: This happens when:

- There are fewer independent equations than variables.
- One or more variables can vary freely.

Step 3: In this traffic loop:

- External flows constrain the system.
- However, one internal circulation flow can vary without violating conservation.

Step 4: Interpretation:

- Multiple internal configurations produce the same observed external data.
- This is common in network flow problems.

Final Answer: A

The measurements constrain the flows, but one internal loop flow is still a free parameter.

—

Q3(C): Controlled Flow Scenario

Question: If the AI controller sets $x_4 = 50$ vehicles/hour, what are the other flows?

Choices:

- A. $(x_1, x_2, x_3, x_4) = (50, 100, 0, 150)$
- B. $(x_1, x_2, x_3, x_4) = (150, 0, 100, 50)$
- C. $(x_1, x_2, x_3, x_4) = (100, 0, 150, 50)$
- D. $(x_1, x_2, x_3, x_4) = (100, 50, 150, 0)$

Solution:

Using the system:

$$x_4 = 50$$

$$x_4 = 50 + x_2 \Rightarrow 50 = 50 + x_2 \Rightarrow x_2 = 0$$

$$x_1 = 100 + x_2 = 100$$

$$x_3 = 50 + x_1 = 150$$

Final Answer: C

$$(x_1, x_2, x_3, x_4) = (100, 0, 150, 50)$$

—

Q3(D): Parametric Solution

Question: In the digital twin, the AI learns the free parameter r such that $x_4 = r$. Which parametric solution matches the model?

Choices:

A. $x_1 = r, x_2 = r, x_3 = r, x_4 = r$

B. $x_1 = 50 - r, x_2 = 50 + r, x_3 = 100 - r, x_4 = r$

C. $x_1 = r + 50, x_2 = r - 50, x_3 = r + 100, x_4 = r$

D. $x_1 = r - 50, x_2 = r + 50, x_3 = r - 100, x_4 = r$

Solution:

Let $x_4 = r$.

From equations:

$$x_4 = 50 + x_2 \Rightarrow x_2 = r - 50$$

$$x_1 = 100 + x_2 = 100 + (r - 50) = r + 50$$

$$x_3 = 50 + x_1 = 50 + (r + 50) = r + 100$$

Thus:

$$x_1 = r + 50, \quad x_2 = r - 50, \quad x_3 = r + 100, \quad x_4 = r$$

Final Answer: C

4 Quiz 2

Question 1

For the system $AX = b$ where:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & -1 & 3 \\ 0 & 2 & 5 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 12 \\ 17 \end{bmatrix}$$

We want to solve $Ax = b$ using LU factorization without pivoting:

$$A = LU, \quad \text{then solve } Ly = b, \quad \text{then } Ux = y.$$

(a) The correct U matrix is:

A. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$

B. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 7 \end{bmatrix}$

C. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$

D. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 3 \\ 0 & 0 & 7 \end{bmatrix}$

Solution:

Perform Gaussian elimination on A :

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_3 \leftarrow R_3$$

$$\Rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 2 & 5 \end{bmatrix}$$

Next eliminate below pivot in column 2:

$$R_3 \leftarrow R_3 - \left(-\frac{2}{3}\right) R_2$$

$$\Rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 7 \end{bmatrix}$$

Correct answer: B

—

(b) The correct L matrix is:

A. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & \frac{2}{3} & 4 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}$

D. $\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix}$

Solution:

From elimination steps:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}$$

Correct answer: C

(c) Solving $Ly = b$, the intermediate vector y equals:

A. $\begin{bmatrix} 3 \\ 0 \\ 13 \end{bmatrix}$

B. $\begin{bmatrix} 3 \\ -6 \\ 21 \end{bmatrix}$

C. $\begin{bmatrix} 3 \\ 12 \\ 17 \end{bmatrix}$

D. $\begin{bmatrix} 3 \\ 6 \\ 21 \end{bmatrix}$

Solution:

Solve $Ly = b$:

$$\begin{aligned}y_1 &= 3 \\2y_1 + y_2 &= 12 \Rightarrow 6 + y_2 = 12 \Rightarrow y_2 = 6 \\-\frac{2}{3}y_2 + y_3 &= 17 \Rightarrow -4 + y_3 = 17 \Rightarrow y_3 = 21\end{aligned}$$

$$y = \begin{bmatrix} 3 \\ 6 \\ 21 \end{bmatrix}$$

Correct answer: D

—

(d) Solving $Ux = y$, the final solution x is:

A. $\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$

C. $\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$

D. $\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$

Solution:

Solve $Ux = y$:

$$\begin{aligned}7x_3 &= 21 \Rightarrow x_3 = 3 \\-3x_2 + 3x_3 &= 6 \Rightarrow -3x_2 + 9 = 6 \Rightarrow x_2 = 1 \\2x_1 + x_2 &= 3 \Rightarrow 2x_1 + 1 = 3 \Rightarrow x_1 = 1\end{aligned}$$

$$x = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

Correct answer: B

5 Quiz 3

Question 1

A scalar λ is an eigenvalue of an $n \times n$ matrix A if there exists a nonzero vector x such that:

- A. $AX = 0$
- B. $AX = \lambda X$
- C. $A^T X = \lambda X$
- D. $(A + \lambda I)X = 0$

Solution:

By definition, λ is an eigenvalue of A if there exists a nonzero vector x such that:

$$Ax = \lambda x$$

This equation means that multiplying x by A only scales the vector by λ .

Option A corresponds to $Ax = 0$, which is only true for $\lambda = 0$ and does not define eigenvalues in general.

Option C involves A^T , which is unrelated unless explicitly stated.

Option D is incorrect because the correct form is $(A - \lambda I)x = 0$.

Correct answer: B

Question 2

The characteristic equation of a matrix A is:

- A. $|A| = 0$
- B. $|A + \lambda I| = 0$
- C. $|A - \lambda I| = 0$
- D. $|A^T - \lambda I| = 0$

Solution:

The characteristic polynomial of a matrix A is defined as:

$$\det(A - \lambda I)$$

The characteristic equation is obtained by setting this determinant equal to zero:

$$\det(A - \lambda I) = 0$$

Option A only gives the determinant of A , not the characteristic equation.

Option B has the wrong sign.

Option D uses A^T , which is unnecessary since A and A^T have the same eigenvalues but the definition is based on A .

Correct answer: C

—

Question 3

If A is diagonalizable and $A = PDP^{-1}$, then A^k equals:

A. $A^k = P^k DP^{-k}$

B. $A^k = PD^k P^{-1}$

C. $A^k = D^k$

D. $A^k = P^{-1} D^k P$

Solution:

Given:

$$A = PDP^{-1}$$

Compute powers:

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PD^2P^{-1}$$

By induction:

$$A^k = PD^k P^{-1}$$

Option A is incorrect because powers do not distribute over matrix products like that.

Option C ignores the similarity transformation.

Option D reverses the order incorrectly.

Correct answer: B

—

Question 4

Use Cayley–Hamilton to reduce a power. Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

Which is correct?

A. $A^2 - 5A + 6I = 0$

B. $A^2 + 5A + 6I = 0$

C. $A^2 - 6A + 5I = 0$

D. $A^2 - 5I = 0$

Solution:

First compute the characteristic polynomial:

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 1 \\ 0 & 3 - \lambda \end{vmatrix} = (2 - \lambda)(3 - \lambda)$$

$$= \lambda^2 - 5\lambda + 6$$

By Cayley–Hamilton:

$$A^2 - 5A + 6I = 0$$

Correct answer: A

—

Question 5

Suppose A is diagonalizable with eigenpairs $\lambda_1 = 2$, $\lambda_2 = 5$ and eigenvectors v_1, v_2 (linearly independent). Which statement is correct?

A. $A^n = (2^n + 5^n)I$

B. $A^n v_1 = n2 v_1$ and $A^n v_2 = n5 v_2$

C. $A^n v_1 = 2^n v_1$ and $A^n v_2 = 5^n v_2$

D. A^n has eigenvalues $2n, 5n$

Solution:

If $Av = \lambda v$, then applying A repeatedly gives:

$$A^2v = A(\lambda v) = \lambda Av = \lambda^2v$$

By induction:

$$A^nv = \lambda^nv$$

Thus:

$$A^nv_1 = 2^nv_1, \quad A^nv_2 = 5^nv_2$$

Option A is incorrect since it assumes A^n is scalar multiple of identity.

Option B incorrectly multiplies by n instead of exponentiating.

Option D is incorrect because eigenvalues become λ^n , not $n\lambda$.

Correct answer: C

—

Question 6

Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

Use Cayley–Hamilton to write A^{2n} in the form $\alpha A + \beta I$, then compute A^4 quickly.

Solution:

Characteristic polynomial:

$$(2 - \lambda)(3 - \lambda) = \lambda^2 - 5\lambda + 6$$

By Cayley–Hamilton:

$$A^2 - 5A + 6I = 0 \quad \Rightarrow \quad A^2 = 5A - 6I$$

Assume:

$$A^n = \alpha_n A + \beta_n I$$

Using eigenvalues:

$$\begin{cases} 2\alpha_n + \beta_n = 2^n \\ 3\alpha_n + \beta_n = 3^n \end{cases}$$

Subtract:

$$\alpha_n = 3^n - 2^n$$

Then:

$$\beta_n = 2^n - 2(3^n - 2^n) = 3 \cdot 2^n - 2 \cdot 3^n$$

Thus:

$$A^n = (3^n - 2^n)A + (3 \cdot 2^n - 2 \cdot 3^n)I$$

Now compute A^4 :

$$A^2 = 5A - 6I$$

$$A^3 = A(5A - 6I) = 5A^2 - 6A = 5(5A - 6I) - 6A = 19A - 30I$$

$$A^4 = A(19A - 30I) = 19A^2 - 30A$$

$$= 19(5A - 6I) - 30A = 95A - 114I - 30A$$

$$A^4 = 65A - 114I$$

$$\boxed{A^4 = 65A - 114I}$$

6 Mid Term

Mid Term PDF

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